
MUHAMMAD ILHAM RAMDHANI

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An enthusiastic undergraduate student in Computer Science at Muhammadiyah University of Cirebon, focusing on software development in the fields of mobile, web, and hardware particularly in the implementation of the Internet of Things (IoT). Possesses strong practical skills in microcontroller programming, front-end and back-end framework development, and automation scripting. Experienced in diagnosing and repairing hardware, with additional initiative in the areas of affiliate community management and market analysis.

Education

Muhammadiyah University of Cirebon

Cirebon, Indonesia

Bachelor of Informatics Engineering.

October 2022 – October 2026

Undergraduate Thesis: Design and Development of an IoT-Based Earthquake Early Warning System using STA/LTA Algorithm and Peak Ground Acceleration (PGA) Verification.

Palimanan State High School No. 1

Cirebon, Indonesia

Natural Science Major (IPA) • Final Score: 94.15/100

May 05, 2022

Academic Honors: 3rd Place in Mathematics at the National Science Olympiad.

Technical & Project Experience

AquaSync (IoT-Based Water Monitoring System)

Cirebon, Indonesia

Independent / Academic Project

July 2026 – August 2026

- Developed an end-to-end IoT architecture utilizing ESP32 microcontrollers and environmental sensors to accurately track and process real-time water quality metrics.
- Designed and deployed a full-stack web dashboard with RESTful APIs, enabling automated data visualization, telemetry logging, and remote actuator control.

Real-Time Hydroponic Monitoring System

Cirebon, Indonesia

Independent IoT Development Project

June 2026 – August 2026

- Developed an automated IoT hydroponic monitoring system to track water pH, temperature, and nutrient levels in real time.
- Implemented microcontrollers and automation scripts to regulate pumps and maintain optimal growth conditions.

Termo App (Thermodynamic Progressive Web App)

Cirebon, Indonesia

Independent IoT Development Project

June 2026 – July 2026

- Engineered a highly responsive Progressive Web Application (PWA) utilizing React.js and Vite, delivering seamless thermodynamic calculations across mobile and desktop environments.
- Integrated Firebase backend services for secure user authentication, cloud data storage, and real-time database synchronization with offline-first capabilities.

IoT-Based Earthquake Early Warning System

Cirebon, Indonesia

Independent / Academic Project

February 2026 – July 2026

- Engineered a multi-stage validation early warning system utilizing STA/LTA algorithms and Peak Ground Acceleration (PGA) verification to detect seismic activities.
- Integrated hardware sensors with microcontrollers and cloud communication to achieve real-time alert processing and minimal detection latency.

IoT Fire Detection System

Cirebon, Indonesia

Fieldwork IoT Development Project

December 2025 – January 2026

- Designed and programmed an IoT-based fire hazard detection system incorporating flame/smoke sensors and automated instant notification mechanisms.

Food-Healthy Web Application

Cirebon, Indonesia

Full-Stack Web Development

November 2025 – December 2025

- Built a web platform for healthy nutrition tracking utilizing front-end and back-end frameworks, implementing structured database designs for efficient user data management.

Leadership & Activities

COMPUTER SCIENCE STUDENT ASSOCIATION

Cirebon, Indonesia

President

November 2022 – December 2024

- Served as Student Body President (2023–2024), leading organizational strategy, student work programs, and external relations for the department.
- Coordinated work plans, executed academic/technical workshops, and managed executive board members across multiple operational divisions.

Art Appreciation Program (PASS)

Cirebon, Indonesia

Band Coordinator

August 2019 – August 2021

- Organized band rehearsals, music showcase schedules, and member talent development initiatives.

Parantapa Bahana Harmoni Marching Band

Cirebon, Indonesia

Field & Training Coordinator

August 2019 – August 2021

- Supervised daily musical and marching drills, structured practice routines, and maintained team performance standards.

GSN ONE Marching Band

Cirebon, Indonesia

Vice President

July 2016 – December 2018

- Assisted in managing organizational operations, member discipline, and event logistics for school competitions and public performances (Period 2017–2018).

Certifications

- Junior Web Developer – National Board for Professional Certification (BNSP) (2025) | No. Reg. TIK 1565 37382 2025
- Basic AI (Belajar Dasar AI) – Dicoding Academy (2024) | dicoding.com/certificates/L4PQ1YROQXO1
- Basic Data Visualization (Belajar Dasar Visualisasi Data) – Dicoding Academy (2024) | dicoding.com/certificates/1RXYL7013PVM

Skills & Interests

Technical Skills: Internet of Things (IoT), Embedded Systems, Microcontroller Programming, Full-Stack Web Development, Artificial Intelligence & Computer Vision, Database Management, Automation Scripting, Cloud Communication Protocols.

Soft Skills & Competencies: Technical Communication, Complex Problem-Solving, Cross-Functional Team Leadership, Project Coordination.

Hardware & Embedded System: C++, Microcontroller Programming, Arduino, ESP8266, ESP32, ESP32-CAM, Robotics, Circuit & Sensor Interfacing.

Web Development: Full-Stack Web Development, HTML, CSS, JavaScript, PHP, Laravel, React, Vite.

Data, AI & Databases: Python, TensorFlow, YOLO (Object Detection), MySQL, Firebase.

Tools & Infrastructure: GitHub, Vercel, Laragon, Postman, MQTT Protocol.

Soft Skills & Competencies: Technical Communication, Complex Problem-Solving, Team Collaboration, Organizational Leadership.

Language: Bahasa Indonesia

Interests: Robotics, Embedded Systems, Internet of Things (IoT) Innovation, Computer Vision & Open-Source Tech.